HW #6

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Reading Homework. Try Exercises 3.1, 3.2, 3.3 in the textbook. Read its solutions in the back.

Question 1. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$. We would like to show that $h_K = 1$.

(1) Use the Minkowski's bound to show that any ideal class has an integral ideal representative $\mathfrak{a}$ with $N(\mathfrak{a}) \leq 3$.

Hint. Use $K = \mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{5})$ to compute $\mathrm{disc}(K)$.

(2) Using how (2) factorizes in $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$, show that the prime ideal factorization of (2) in K is
$$(2) = \mathfrak{q}_2^2,$$
for a unique prime ideal $\mathfrak{q}_2 \subset \mathcal{O}_K$ lying over 2. Deduce that $N(\mathfrak{q}_2) = 4$.

(3) Using how (3) factorizes in $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$, show that 3 does not split completely in K. Deduce that there is no integral prime ideal of $\mathcal{O}_K$ with norm exactly equal to 3. Deduce that $h_K = 1$.

(4) In Exercise 3.2(3), it is shown that there is only one prime ideal $\mathfrak{p}_2$ of $\mathbb{Q}(\sqrt{10})$ dividing 2, and that $\mathfrak{p}_2$ is not a principal ideal. Show that this ideal generates a principal ideal over $\mathbb{Q}(\sqrt{2}, \sqrt{5})$; namely, show that $\mathfrak{p}_2\mathcal{O}_{\mathbb{Q}(\sqrt{2},\sqrt{5})}$ is a principal ideal.[1]

Proof. (1) Note that there are four real embeddings, so the Minkowski bound is $\frac{4!}{4^4}\sqrt{\mathrm{disc}(K)} = \frac{3}{32}\sqrt{\mathrm{disc}(K)}$. Note that $\mathrm{disc}(\mathbb{Q}(\sqrt{2})) = 8$ and $\mathrm{disc}(\mathbb{Q}(\sqrt{5})) = 5$, so they are coprime, so $\mathrm{disc}(K) = 8^2 5^2 = 2^6 \cdot 5^2$, so the Minkowski bound is $\frac{3}{32}2^3 \cdot 5 = \frac{15}{4}$.

(2) Note that (2) factorizes in $\mathbb{Q}(\sqrt{2})$ as $(2) = (2, \sqrt{2})^2 = (\sqrt{2})^2$ and in $\mathbb{Q}(\sqrt{5})$ as $(2) = (2)$. For K and (2), the e, f, g should satisfy $efg = 4$, and we know as (2) is inert in $\mathbb{Q}(\sqrt{5})$ that $f \geq 2$ and as (2) is totally ramified in $\mathbb{Q}(\sqrt{2})$ that $e \geq 2$. So $e = 2$, $f = 2$, $g = 1$. Thus $(2) = \mathfrak{p}_2^2$, with $N(\mathfrak{p}_2) = 2^f = 4$.

(3) Note that (3) factorizes in $\mathbb{Q}(\sqrt{2})$ as $(3) = (3)$, and in $\mathbb{Q}(\sqrt{5})$ as $(3) = (3)$. So $f \geq 2$, so it is not split completely. Thus there is no integral ideal of norm 3. Thus the nontrivial ideal class, if there is any, is represented by $\mathfrak{p}_2$, which is also principal, so $\mathrm{Cl}(K) = 1$.

(4) Note that the prime ideal $\mathfrak{p}_2$ from Exercise 3.2 is $(2, \sqrt{10} + 2) = (2, \sqrt{10})$ in $\mathcal{O}_{\mathbb{Q}(\sqrt{10})}$. In $\mathcal{O}_K$, $(2, \sqrt{10}) \subset (\sqrt{2})$. On the other hand, $\sqrt{5} \cdot \sqrt{10} - (2\sqrt{2}) \cdot 2 = 5\sqrt{2} - 4\sqrt{2} = \sqrt{2}$, so $(\sqrt{2}) \subset (2, \sqrt{10})$, so $(2, \sqrt{10})$ is principal in $\mathcal{O}_K$.

□

Question 2. Let $K = \mathbb{Q}(\sqrt{-23})$, and let $p \neq 23$ be a rational prime.

[1] In fact, $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ is the **Hilbert class field** of $\mathbb{Q}(\sqrt{10})$ (one related fact is that $[\mathbb{Q}(\sqrt{2}, \sqrt{5}) : \mathbb{Q}(\sqrt{10})] = 2 = h_{\mathbb{Q}(\sqrt{10})}$); we will see later that this is one of the properties of the Hilbert class field, that any prime ideal downstairs becomes principal in the Hilbert class field.

(1) Use the Minkowski's bound to show that any ideal class has an integral ideal representative $\mathfrak{a}$ with $N(\mathfrak{a}) \leq 3$.

(2) Using the factorization of the principal ideals (2), (3), $\left(\frac{1+\sqrt{-23}}{2}\right)$ and $\left(\frac{3+\sqrt{-23}}{2}\right)$, show that $h_K = 3$, with

$$\mathrm{Cl}(K) = \left\{[(1)], \left[\left(2, \frac{1+\sqrt{-23}}{2}\right)\right], \left[\left(2, \frac{-1+\sqrt{-23}}{2}\right)\right]\right\}.$$

(3) Show that p is properly represented by either $X^2 - XY + 6Y^2$, $2X^2 + XY + 3Y^2$ or $2X^2 - XY + 3Y^2$ if and only if -23 is a square mod p.

(4) Show that $p = 2X^2 \pm XY + 3Y^2$ for some $X, Y \in \mathbb{Z}$ if and only if $2p = Z^2 - ZW + 6W^2$ for some $Z, W \in \mathbb{Z}$.

(5) Combining the above, show that, for $p \neq 23$.

$$X^2 - XY + 6Y^2 \text{ represents either } p \text{ or } 2p \Leftrightarrow p \text{ is a square mod } 23.$$

(6) Show that the two cases in the left side of (5) are mutually exclusive. Namely, show that there is no $p \neq 23$ such that $X^2 - XY + 6Y^2$ represents both p and $2p$.

(7) Show that, for $p \neq 23$.

$$X^2 - XY + 6Y^2 \text{ represents either } p \text{ or } 2p \Leftrightarrow X^2 + 23Y^2 \text{ represents either } 4p \text{ or } 8p.$$

(8) Show that, for $p \neq 23$,

$$X^2 + 23Y^2 \text{ represents } 4p \Leftrightarrow X^2 + 23Y^2 \text{ represents } p.$$

Hint. $n^2 \equiv 0, 1, 4 \pmod 8$.

(9) Deduce that

$$X^2 + 23Y^2 \text{ represents either } p \text{ or } 8p \Leftrightarrow p \text{ is a square mod } 23.$$

Is (8) true with $8p$ and $2p$? Namely, does $X^2 + 23Y^2$ represent $8p$ if and only if $X^2 + 23Y^2$ represents $2p$?

Proof. (1) The Minkowski's bound for K is

$$B_K = \frac{2}{4}\frac{4}{\pi}\sqrt{23} \sim 3.05$$

which shows what the problem asked.

(2) Note that $\mathcal{O}_K \cong \mathbb{Z}[X]/(X^2 - X + 6)$. As $X^2 - X + 6 \equiv X(X-1)$ modulo 2 or 3, this implies that

$$(2) = \left(2, \frac{1+\sqrt{-23}}{2}\right)\left(2, \frac{-1+\sqrt{-23}}{2}\right)$$

$$(3) = \left(3, \frac{1+\sqrt{-23}}{2}\right)\left(3, \frac{-1+\sqrt{-23}}{2}\right)$$

Let us name the primes $\mathfrak{p}_2, \mathfrak{p}_2', \mathfrak{p}_3, \mathfrak{p}_3'$ so that the prime ideal factorization above is $(2) = \mathfrak{p}_2\mathfrak{p}_2'$ and $(3) = \mathfrak{p}_3\mathfrak{p}_3'$ in order. Then, by (1), any element in $\mathrm{Cl}(K)$ is either $[(1)]$, $[\mathfrak{p}_2]$, $[\mathfrak{p}_2']$,

$[\mathfrak{p}_3]$, or $[\mathfrak{p}_3']$. Moreover, $[\mathfrak{p}_2'] = [\mathfrak{p}_2]^{-1}$, and $[\mathfrak{p}_3'] = [\mathfrak{p}_3]^{-1}$. Note that $N_{K/\mathbb{Q}}\left(\frac{1+\sqrt{-23}}{2}\right) = \frac{1+23}{4} = 6$, and as $\mathfrak{p}_2, \mathfrak{p}_3 | \left(\frac{1+\sqrt{-23}}{2}\right)$, it follows that

$$\left(\frac{1+\sqrt{-23}}{2}\right) = \mathfrak{p}_2\mathfrak{p}_3.$$

Therefore, in $\mathrm{Cl}(K)$, $[\mathfrak{p}_3] = [\mathfrak{p}_2]^{-1} = [\mathfrak{p}_2']$, which implies that $[\mathfrak{p}_3'] = [\mathfrak{p}_2]$, and therefore $\#\,\mathrm{Cl}(K) \leq 3$. The desired conclusion will follow if we show that $[\mathfrak{p}_2]$ is exactly of order 3; i.e., if we show that $\mathfrak{p}_2$ is not principal and $\mathfrak{p}_2^3$ is principal.

It is easy to show that $\mathfrak{p}_2$ is not principal, as there is no $\alpha \in \mathcal{O}_K$ with $N_{K/\mathbb{Q}}(\alpha) = \pm 2$; any $\alpha \in \mathcal{O}_K$ is of the form $\frac{x+y\sqrt{-23}}{2}$ with $x, y \in \mathbb{Z}$, so $N_{K/\mathbb{Q}}(\alpha) = \frac{x^2+23y^2}{4}$ which cannot be ± 2 as $x^2 + 23y^2 \neq \pm 8$.

Note that $N_{K/\mathbb{Q}}\left(\frac{3+\sqrt{-23}}{2}\right) = \frac{9+23}{4} = \frac{32}{4} = 8$, so $\left(\frac{3+\sqrt{-23}}{2}\right)$ factors as a product of three prime ideals dividing 2. Thus, it can be either $\mathfrak{p}_2^3$, $\mathfrak{p}_2^2\mathfrak{p}_2' = 2\mathfrak{p}_2$, $\mathfrak{p}_2\mathfrak{p}_2'^2 = 2\mathfrak{p}_2'$, or $\mathfrak{p}_2'^3$. On the other hand, as $\frac{3+\sqrt{-23}}{2}$ is not divisible by 2 in $\mathcal{O}_K$, so the only possibilities are $\mathfrak{p}_2^3$ or $\mathfrak{p}_2'^3$. Note that $\frac{3+\sqrt{-23}}{2} \in \left(2, \frac{-1+\sqrt{-23}}{2}\right)$. Therefore, $\mathfrak{p}_2'$ divides $\frac{3+\sqrt{-23}}{2}$; this implies that $\frac{3+\sqrt{-23}}{2} = \mathfrak{p}_2'^3$. This implies that $\mathfrak{p}_2'^3$ is principal, so $\mathfrak{p}_2^3$ is principal as well.

(3) We want to use Theorem 3.23. Note that $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{-23}}{2}\right] = \mathbb{Z} + \mathbb{Z}\frac{1+\sqrt{-23}}{2}$, so under the bijection of Theorem 3.23, it should correspond to $X^2 - XY + 6Y^2$ (its discriminant is $(-1)^2 - 4 \cdot 6 = -23$). Similarly, as

$$\left(2, \frac{\pm 1+\sqrt{-23}}{2}\right) = 2\mathbb{Z} + \mathbb{Z}\frac{\pm 1+\sqrt{-23}}{2},$$

this is equivalent as an ideal class to $\frac{1}{2}\left(2, \frac{\pm 1+\sqrt{-23}}{2}\right) = \mathbb{Z} + \mathbb{Z}\frac{\pm 1+\sqrt{-23}}{4}$. Thus, under the bijection of Theorem 3.23, it should correspond to $2X^2 \mp XY + 3Y^2$. Now the desired statement is a consequence of Proposition 3.20.

(4) If $p = 2X^2 \pm XY + 3Y^2$, then

$$2p = 4X^2 \pm 2XY + 6Y^2 = (2X \pm Y)^2 - (2X \pm Y)(\pm Y) + 6(\pm Y)^2.$$

Conversely, if $2p = Z^2 - ZW + 6W^2$, $Z^2 \equiv ZW \pmod 2$, so either Z is even, or $Z \equiv W \pmod 2$. If Z is even, then $Z = 2A$, and we have $2p = 4A^2 - 2AW + 6W^2$, or $p = 2A^2 - AW + 3W^2$. If $Z \equiv W \pmod 2$, then $Z = W + 2B$ for some $B \in 2\mathbb{Z}$, and we have

$$2p = (W+2B)^2 - (W+2B)W + 6W^2 = 4B^2 + 2WB + 6W^2,$$

or $p = 2B^2 + WB + 3W^2$. Either way, $p = 2X^2 \pm XY + 3Y^2$ for some $X, Y \in \mathbb{Z}$.

(5) This follows from (3) and (4) and $\left(\frac{-23}{p}\right) = \left(\frac{p}{23}\right)$ which is a consequence of the quadratic reciprocity.

(6) Note that $N_{K/\mathbb{Q}}\left(X - Y\frac{1+\sqrt{-23}}{2}\right) = X^2 - XY + 6Y^2$. Therefore, if p and $2p$ are represented by $X^2 - XY + 6Y^2$, there are $\alpha, \beta \in \mathcal{O}_K$ such that $N_{K/\mathbb{Q}}(\alpha) = p$ and $N_{K/\mathbb{Q}}(\beta) = 2p$. This implies that $\mathfrak{p} := (\alpha)$ is a prime ideal of $\mathcal{O}_K$ dividing p, and because of the existence of $\mathfrak{p}$, (p) factors in K as $(p) = \mathfrak{p}\mathfrak{p}'$ for a prime ideal $\mathfrak{p}' \neq \mathfrak{p}$ of ideal norm p. Note

that $[\mathfrak{p}'] = [(p)][\mathfrak{p}]^{-1} = 1$ in $\mathrm{Cl}(K)$. On the other hand, (β) should factor as a product of a prime ideal of norm 2 and a prime ideal of norm p. As both prime ideals of norm p are principal, the prime ideal factor of (β) of norm 2 should be principal. However, this contradicts our earlier observation in (2). This implies that $X^2 - XY + 6Y^2$ cannot represent both p and $2p$.

(7) Note that $4(X^2 - XY + 6Y^2) = (2X - Y)^2 + 23Y^2$, so the $\Rightarrow$ direction is clear. Conversely, suppose that $X^2 + 23Y^2$ represents either $4p$ or $8p$. Then, $X^2 + 23Y^2$ is divisible by 4, so $X^2 \equiv Y^2 \pmod 4$, which implies that $X \equiv Y \pmod 2$. Writing $X = 2A - Y$, we have

$$X^2 + 23Y^2 = 4A^2 - 4AY + 24Y^2.$$

Thus $A^2 - AY + 6Y^2$ is either p or $2p$, as desired.

(8) The $\Leftarrow$ direction is clear. Conversely, if $4p = X^2 + 23Y^2$, then $X \equiv Y \pmod 2$ as observed above. If X and Y are both odd, then $X^2 + 23Y^2 \equiv 1 + 23 \equiv 0 \pmod 8$, so p is even. As p is a prime, $p = 2$, but it is impossible to have $8 = X^2 + 23Y^2$. Therefore, X, Y are both even, and we get the $\Rightarrow$ direction.

(9) The first statement follows from (5), (7), (8). The analogue of (8) is false for $8p$ and $2p$; for $p = 3$, $X^2 + 23Y^2 = 24$ for $X = Y = 1$, but $X^2 + 23Y^2 \neq 6$ for any $X, Y \in \mathbb{Z}$.

$\square$